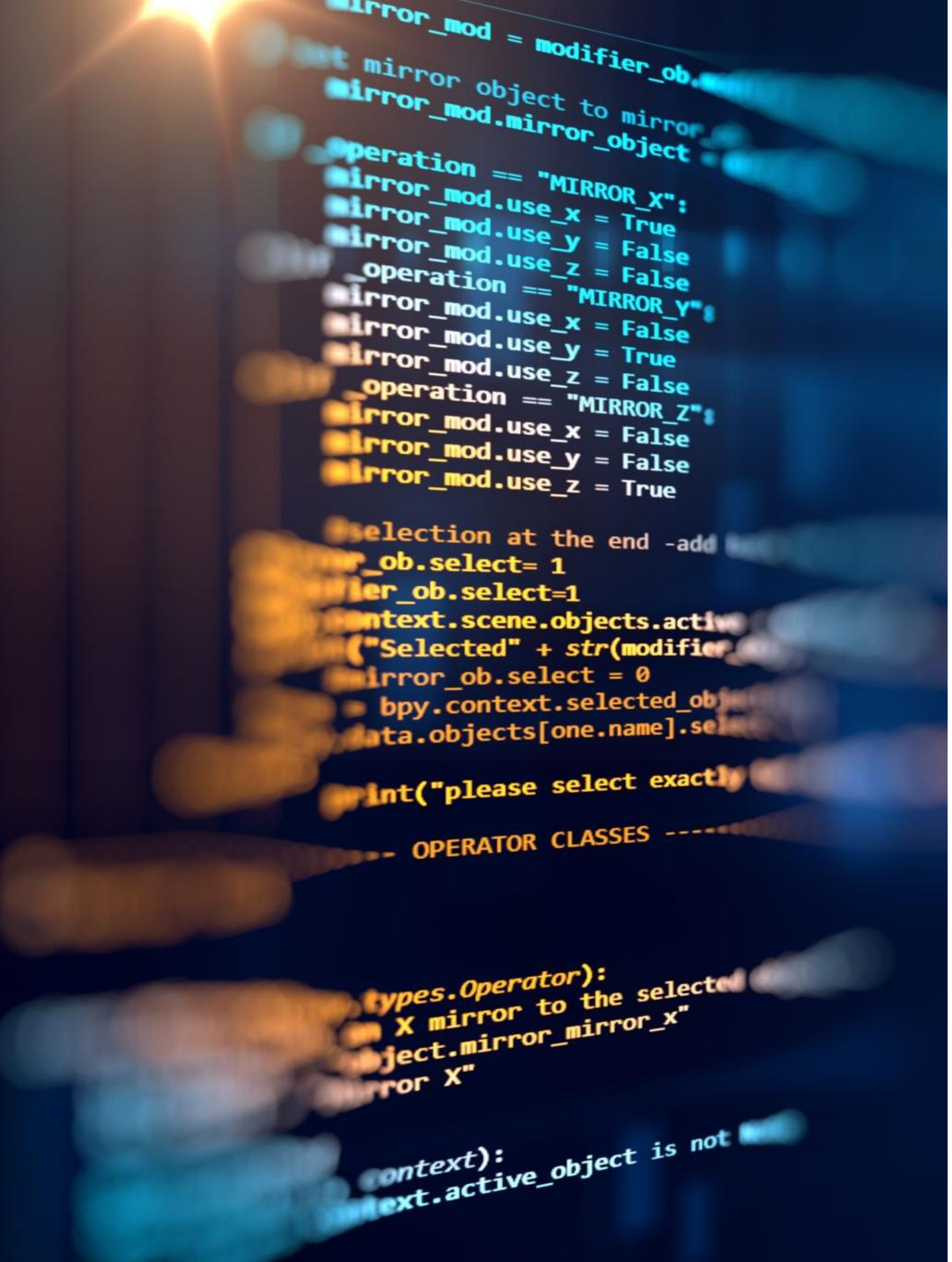
Basics of Cyber-Security Networking

Networking principles: OSI model, MAC and IP addressing



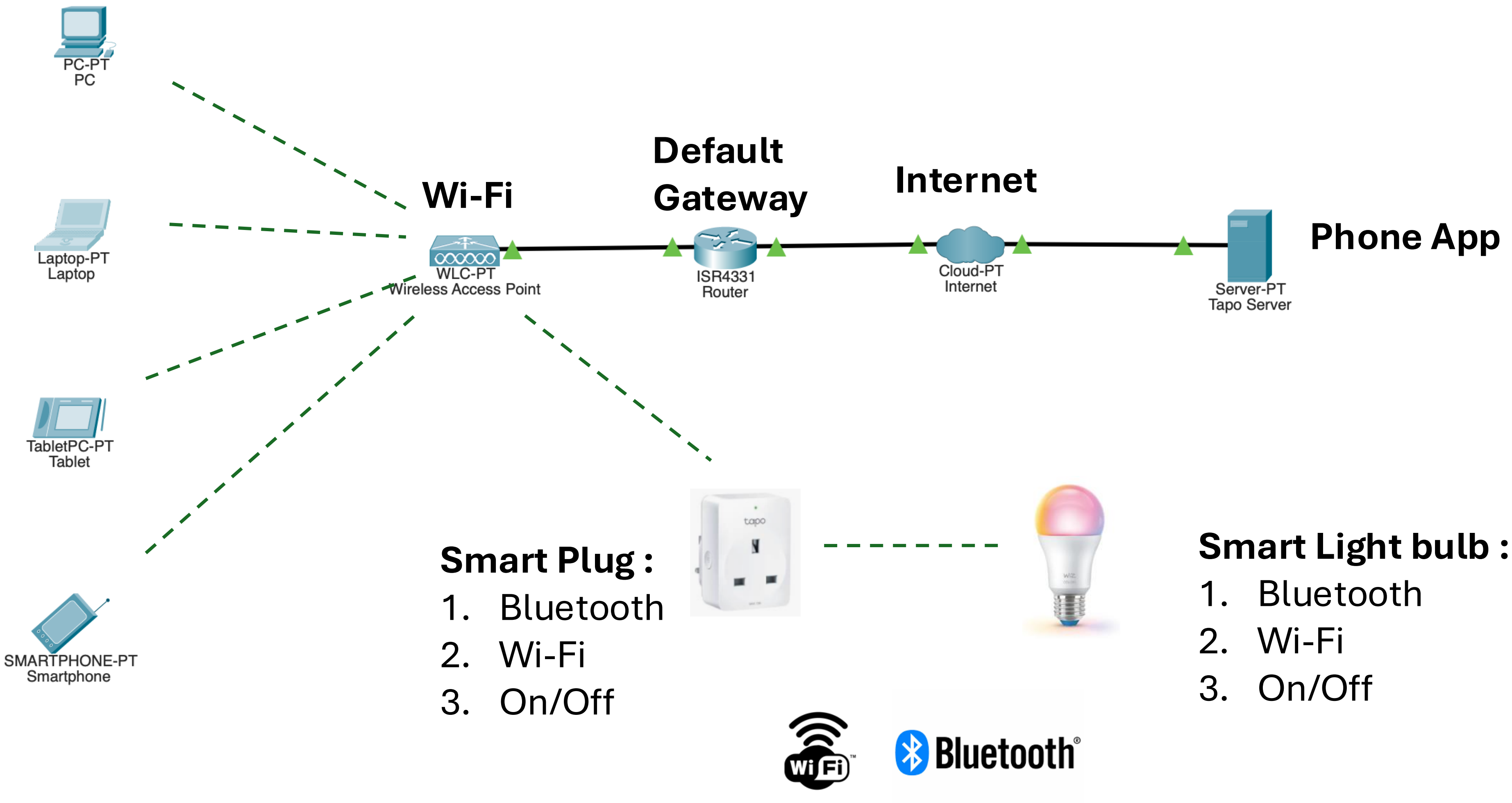
Ethical Hacking Statement

- A hacker is a person who **breaks into a computer system**.
- The reasons for hacking can be many:
  - installing malware, stealing or destroying data, disrupting service, and more.
- Hacking can also be done for **ethical reasons**, such as trying to find software vulnerabilities so they can be fixed.
- Ethical hacking involves the **legal use of hacking techniques** for benevolent versus malicious purposes.
- Ethical hackers use **penetration testing** and other tactics to find software vulnerabilities and other security weaknesses so they can be promptly addressed.
  - **Permission is required**
- https://www.cisco.com/c/en/us/products/security/what-is-a-hacker.html

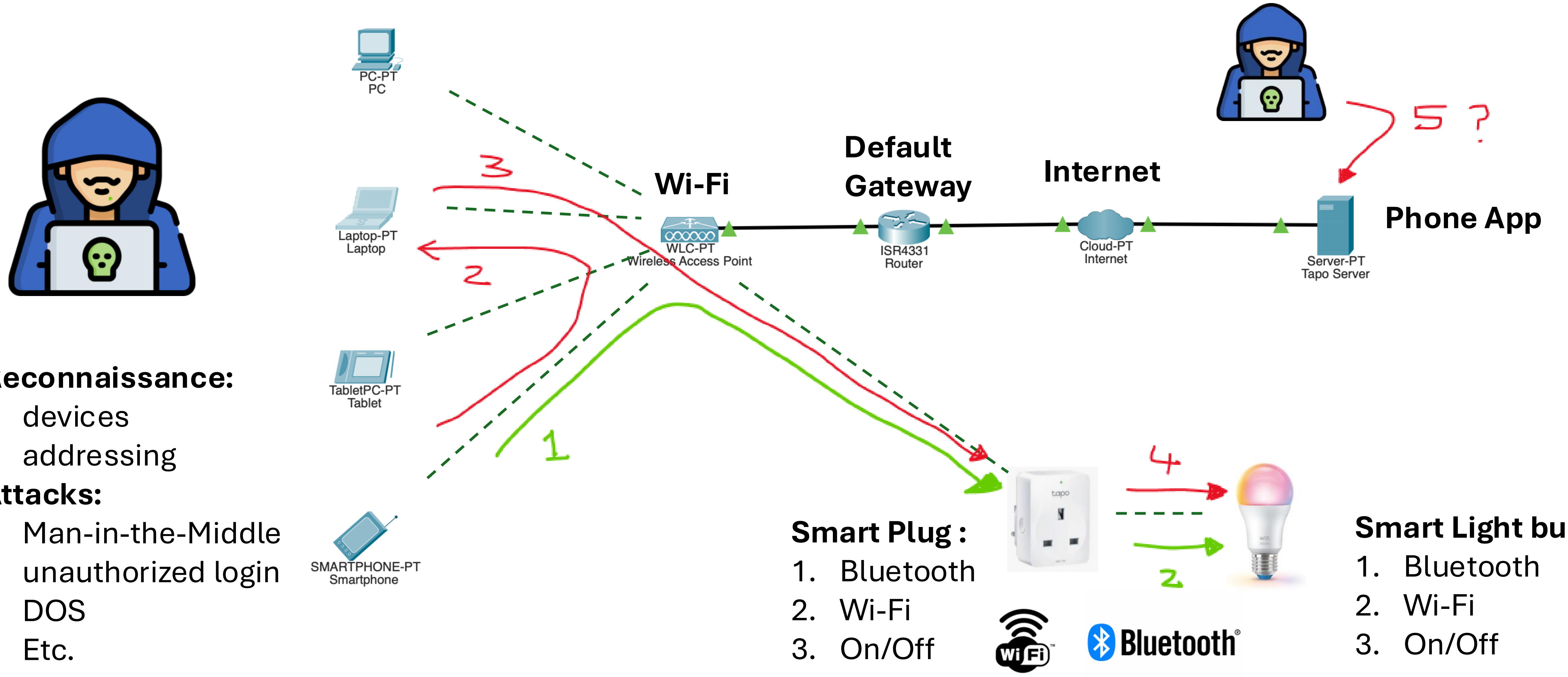
The Network

Connectivity:

1. Bluetooth
2. Wi-fi
3. Internet



Hacker



Reconnaissance:
- devices
- addressing

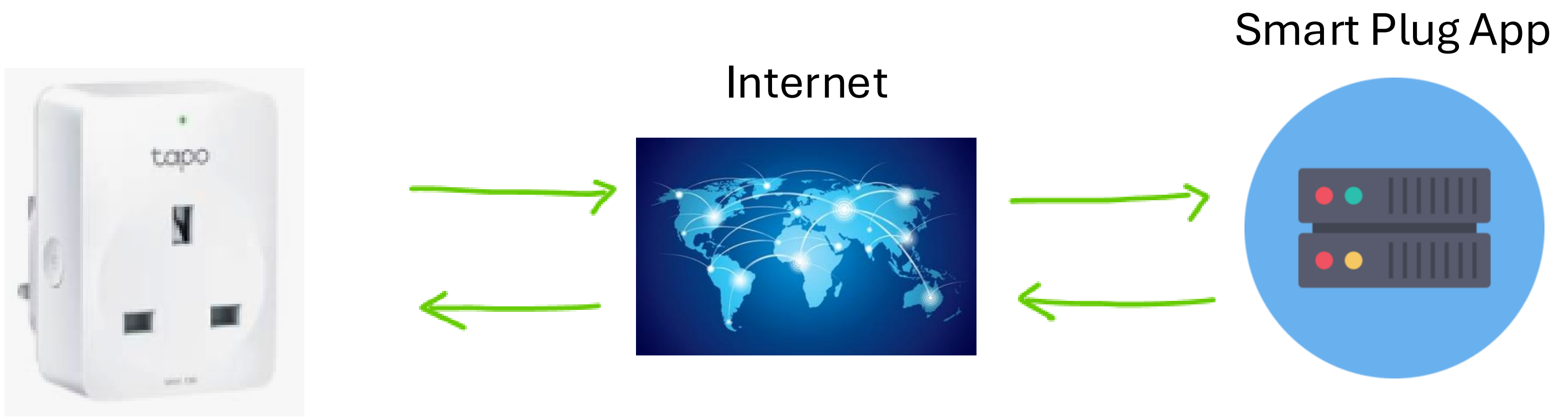
Attacks:
- Man-in-the-Middle
- unauthorized login
- DOS
- Etc.

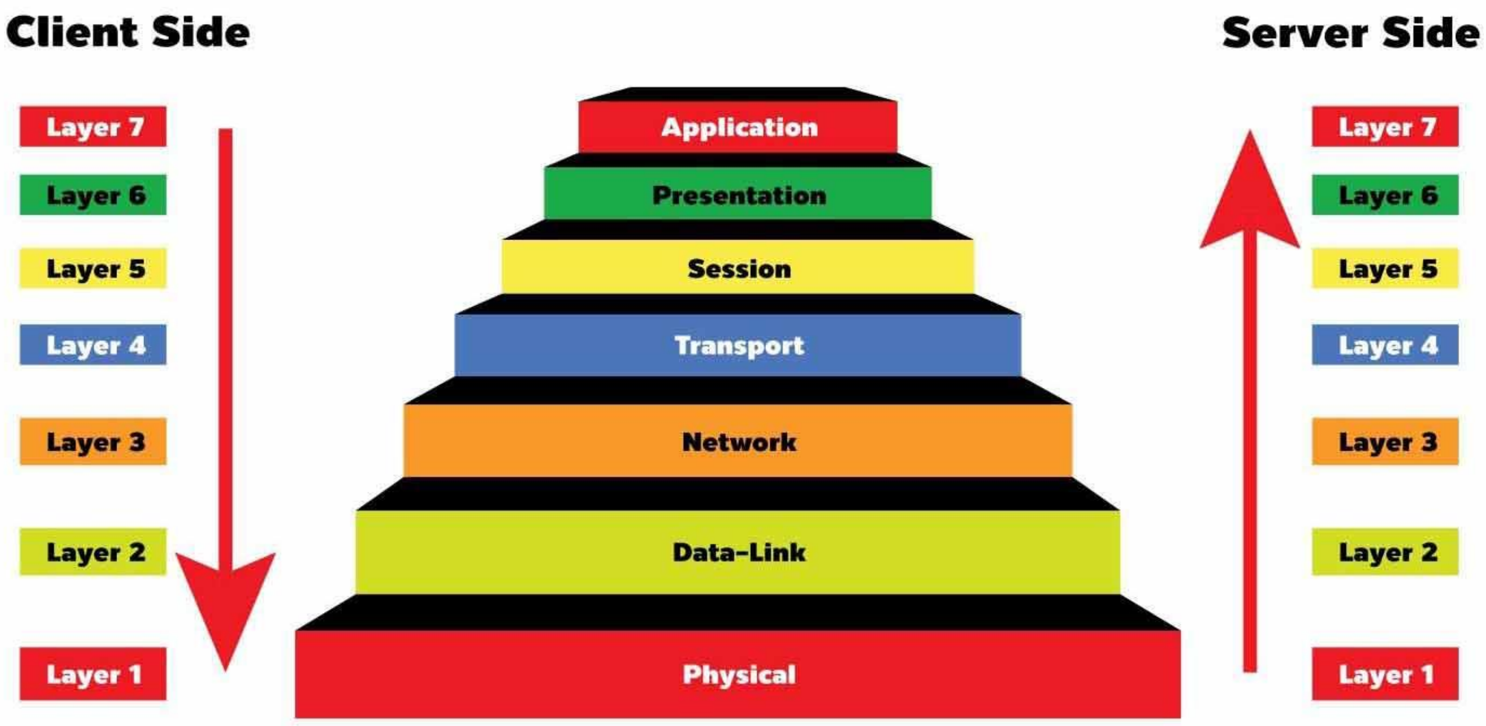
Smart Plug :
1. Bluetooth
2. Wi-Fi
3. On/Off

Smart Light bulb :
1. Bluetooth
2. Wi-Fi
3. On/Off

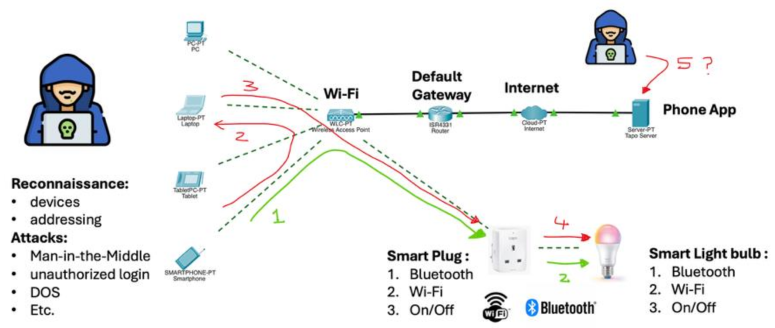
OSI model

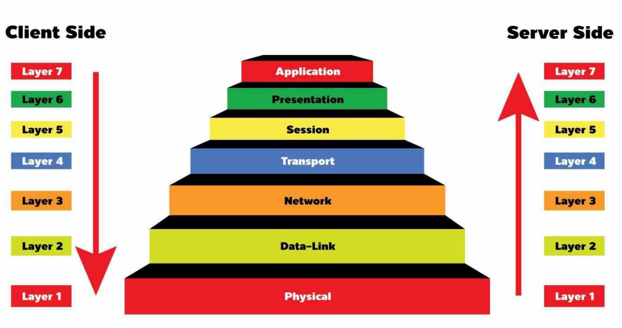
describes seven layers that computer systems use to communicate over a network





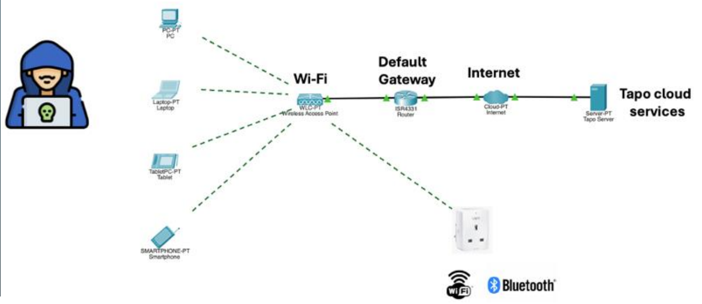
Addressing





OSI Layer	Protocol	Detail	Example
Layer 5-7 (Application)	Data	Application	e.g. Smart plug app
Layer 4 (Transport)	TCP or UDP	Port number	e.g. 443, 8009 etc
Layer 3 (Network)	IPv4 or IPv6	IP addressing	e.g. 192.168.1.10 (32 bits) e.g. 2001:0000:130F:0000:0000:09C0:876A:130B (128 bits)
Layer 2 (Data Link)	Wi-Fi	MAC addressing	e.g. 00-B0-D0-63-C2-26 (48 bits, 12 Hexadecimal)
	Wi-Fi	SSID	*IPL Wifi*
	Bluetooth	MAC addressing	e.g. 00:11:22:33:FF:EE (48 bits, 12 Hexadecimal)
Layer 1 (Physical)	Wi-Fi & Bluetooth	Radio Frequency	2.4 and 5 GHz bands

The Hacking challenges…



OSI Layer	Protocol	Find Smart plug, bulb, app…	Hack Success?
Layer 5-7	Application	Authentication?	✓ or ✗
Layer 4	TCP or UDP	Port number?	✓ or ✗
Layer 3	IPv4 or IPv6	IP address?	✓ or ✗
Layer 2	Wi-Fi / Bluetooth	MAC address?	✓ or ✗
	Wi-Fi / Bluetooth	Authentication?	IPL wi-fi ✓
	Wi-Fi / Bluetooth	SSID?	IPL wi-fi ✓
Layer 1	Wi-Fi / Bluetooth	Frequency?	IPL wi-fi ✓

Enjoy!